ETSI TS 102 870-2 V1.2.1 (2022-03)

Intelligent Transport Systems (ITS);
Testing;
Conformance test specifications for GeoNetworking Basic Transport Protocol (BTP);
Part 2: Test Suite Structure and Test Purposes (TSS&TP)

Reference
RTS/ITS-00367

Keywords
ITS, testing, transport, TSS&TP

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

The present document is part 2 of a multi-part deliverable covering Conformance test specification for Geonetworking Basic Transport Protocol (BTP**)** as identified below:

Part 1: "Test requirements and Protocol Implementation Conformance Statement (PICS) pro forma";

Part 2: "Test Suite Structure and Test Purposes (TSS&TP)";

Part 3: "Abstract Test Suite (ATS) and Protocol Implementation eXtra Information for Testing (PIXIT)".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Test Suite Structure and Test Purposes (TSS&TP) for Geonetworking Basic Transport Protocol (BTP) as defined in ETSI EN 302 636-5-1 [1] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.5].

The ISO standard for the methodology of conformance testing (ISO/IEC 9646-1 [i.2] and ISO/IEC 9646-2 [i.3]) as well as the ETSI rules for conformance testing (ETSI ETS 300 406 [i.6]) are used as a basis for the test methodology.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 302 636-5-1 (V2.2.1): "Intelligent Transport Systems (ITS); Vehicular Communications; GeoNetworking; Part 5: Transport Protocols; Sub-part 1: Basic Transport Protocol".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI EG 202 798: "Intelligent Transport Systems (ITS); Testing; Framework for conformance and interoperability testing".

[i.2] ISO/IEC 9646-1 (1994): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[i.3] ISO/IEC 9646-2 (1994): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 2: Abstract Test Suite specification".

[i.4] ISO/IEC 9646-6 (1994): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 6: Protocol profile test specification".

[i.5] ISO/IEC 9646-7 (1995): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

[i.6] ETSI ETS 300 406 (1995): "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI EN 302 636-5-1 [1], ISO/IEC 9646-6 [i.4] and ISO/IEC 9646-7 [i.5] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

BI	Invalid Behaviour
BTP	Basic Transport Protocol
BV	Valid Behaviour
ITS	Intelligent Transportation Systems
IUT	Implementation Under Test
TP	Test Purposes
TSS	Test Suite Structure

4 Test Suite Structure (TSS)

4.1 Structure for BTP tests

Table 1 shows the BTP Test Suite Structure (TSS) including its subgroups defined for conformance testing.

Table 1

Root	Group	Category
BTP	Packet Generation BTP-A	Valid behaviour
	Packet Generation BTP-B	Valid behaviour
	Packet Processing	Valid behaviour

4.2 Test groups

4.2.1 Introduction

The test suite has a total of three levels. The first level is the root. The second level separates the root into various functional areas. The third level is the standard ISO conformance test categories.

4.2.2 Root

The root identifies the Geonetworking Basic Transport Protocol (BTP**)** given in ETSI EN 302 636-5-1 [1].

4.2.3 Groups

This level contains three functional areas identified as: Packet Generation BTP-A, Packet Generation BTP-B and Packet processing.

4.2.4 Categories

This level contains the standard ISO conformance test categories limited to the valid behaviour and the invalid behaviour.

5 Test Purposes (TP)

5.1 Introduction

5.1.1 TP definition conventions

The TPs are defined by the rules shown in table 2.

Table 2: TP definition rules

TP Header	
TP ID	The TP ID is a unique identifier. It shall be specified according to the TP naming conventions defined in clause 5.1.2.
Test objective	Short description of test purpose objective according to the requirements from the base standard.
Reference	The reference indicates the sub-clauses of the reference standard specifications in which the conformance requirement is expressed.
PICS Selection	Reference to the PICS statement involved for selection of the TP. Contains a Boolean expression.
TP Behaviour	
Initial conditions	The initial conditions defines in which initial state the IUT has to be to apply the actual TP. In the corresponding Test Case, when the execution of the initial condition does not succeed, it leads to the assignment of an Inconclusive verdict.
Expected behaviour(TP body)	Definition of the events, which are parts of the TP objective, and the IUT are expected to perform in order to conform to the base specification. In the corresponding Test Case, Pass or Fail verdicts can be assigned there.
Final conditions	Definition of the events that the IUT is expected to perform or shall not perform, according to the base standard and following the correct execution of the actions in the expected behaviour above. In the corresponding Test Case, the execution of the final conditions is evaluated for the assignment of the final verdict.

5.1.2 TP Identifier naming conventions

The identifier of the TP is built according to table 3.

Table 3: TP naming convention

Identifier:	**TP/<root>/<gr>/<x>/<nn>**		
	<root> = root	BTP	Basic Transport Protocol
	<gr> = group	PGA	Packet Generation BTP-A
		PGB	Packet Generation BTP-B
		PP	Packet Processing
	<x> = type of testing	BV	Valid Behavior tests
		BI	Invalid Syntax or Behavior Tests
	<nn> = sequential number		01 to 99

5.1.3 Rules for the behaviour description

The description of the TP is built according to ETSI EG 202 798 [i.1].

5.1.4 Sources of TP definitions

All TPs shall be as specified according to ETSI EN 302 636-5-1 [1].

5.2 Test purposes for BTP

5.2.1 Packet Generation

5.2.1.1 BTP-A

TP Id	TP/BTP/PGA/BV/01
Test objective	Checks that BTP-A packet is well-formatted
Reference	ETSI EN 302 636-5-1 [1], clauses 6, 7.1, 7.2.1 and 8.2
PICS Selection	
Initial conditions	

with {
the IUT being in the "initial state"
}	

Expected behaviour	

ensure that {
when {
the IUT is requested to send a BTP packet via a BTP-data request
containing BTP Type
indicating value 'BTP-A'
containing Source Port
indicating value 'SOURCE_PORT'
containing Destination Port
indicating value 'DESTINATION_PORT'
}
then {
the IUT sends a valid BTP-A packet
containing source port
indicating 'SOURCE_PORT',
containing destination port
indicating 'DESTINATION_PORT',
containing the Upper Layer payload
}
}	

5.2.1.2 BTP-B

TP Id	TP/BTP/PGB/BV/01
Test objective	Checks that BTP-B packet is well-formatted if Destination Port info is provided
Reference	ETSI EN 302 636-5-1 [1], clauses 6, 7.1, 7.3.2 and 8.2
PICS Selection	
Initial conditions	

with {
the IUT being in the "initial state"
}	

Expected behaviour	

ensure that {
when {
the IUT is requested to send a BTP packet via a BTP-data request
containing BTP Type
indicating value 'BTP-B'
containing Destination Port
indicating value 'DESTINATION_PORT'
containing Destination Port Info
indicating value 'DESTINATION_PORT_INFO'
}
then {
the IUT sends a valid BTP-B packet
containing Destination port
indicating 'DESTINATION_PORT',
containing destination port info
indicating 'DESTINATION_PORT_INFO',
containing the Upper Layer payload
}
}	

TP Id	TP/BTP/PGB/BV/02
Test objective	Checks that BTP-B packet is well-formatted if no Destination Port Info is provided
Reference	ETSI EN 302 636-5-1 [1], clauses 6, 7.1, 7.3.2 and 8.2
PICS Selection	
Initial conditions	

with {
the IUT being in the "initial state"
}	

Expected behaviour	

ensure that {
when {
the IUT is requested to send a BTP packet via a BTP-data request via a BTP-data request
containing BTP Type
indicating value 'BTP-B'
containing Destination Port
indicating value 'DESTINATION_PORT'
not containing Destination Port Info parameter
}
then {
the IUT sends a valid BTP-B packet
containing destination port
indicating 'DESTINATION_PORT',
containing destination port info
indicating '0',
containing the Upper Layer payload
}
}	

5.2.3 Packet processing

5.2.3.1 Valid

TP Id	TP/BTP/PP/BV/01
Test objective	Checks that BTP passes a valid BTP-A packet to the upper protocol entity
Reference	ETSI EN 302 636-5-1 [1], clauses 6, 7.1, 7.2.1 and 8.3
PICS Selection	
Initial conditions	

with {
the IUT being in the "initial state"
}	

Expected behaviour	

ensure that {
when {
the IUT receives a valid BTP-A packet
containing source port
indicating 'SOURCE_PORT',
containing destination port
indicating 'DESTINATION_PORT',
containing payload
}
then {
the IUT passes the payload to the upper layer
}
}	

TP Id	TP/BTP/PP/BV/02
Test objective	Checks that BTP passes a valid BTP-B packet to the upper protocol entity
Reference	ETSI EN 302 636-5-1 [1], clauses 6, 7.1, 7.3.1 and 8.3
PICS Selection	
Initial conditions	

with {
the IUT being in the "initial state"
}	

Expected behaviour	

ensure that {
when {
the IUT receives a valid BTP-B packet
containing destination port
indicating 'DESTINATION_PORT',
containing destination port info
indicating 'DESTINATION_PORT_INFO'
containing payload
}
then {
the IUT passes the payload to the upper protocol
}
}	

Annex A (informative): Bibliography

ETSI TS 102 870-1 (V1.2.1): "Intelligent Transport Systems (ITS); Testing; Conformance test specifications for Geonetworking Basic Transport Protocol (BTP); Part 1: Test requirements and Protocol Implementation Conformance Statement (PICS) pro forma".

"GeoNet STREP N 216269 - D2.2 Final GeoNet Specification".

History

Document history		
V1.1.1	March 2011	Publication
V1.2.1	March 2022	Publication